

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12408456

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Li; Cheng-Yen et al.

Image sensor pixel and metal shielding of charge storage device of image sensor pixel formed by one step process

Abstract

A method is provided for light shielding a charge storage device of an image sensor pixel that includes a photosensitive device and the charge storage device and a dielectric layer covering the photosensitive device and the charge storage device. The method includes performing etching of the dielectric layer to define an undercut volume beneath the dielectric layer and an access opening through the dielectric layer to the undercut volume, and performing physical vapor deposition (PVD) of a light blocking material to both: fill the undercut volume with the light blocking material to form a light blocking layer covering the charge storage device, and fill the access opening with the light blocking material to form a light blocking plug. An image sensor pixel formed by such a process, and an image sensor comprising an array of image sensor pixels, are also disclosed.

Inventors: Li; Cheng-Yen (Hsinchu, TW), Chen; Chia-Chan (Zhubei, TW), Lee; Meng-Chin (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 82762098

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Appl. No.: 18/213320

Filed: June 23, 2023

Prior Publication Data

Document Identifier

US 20230335569 A1

Publication Date

Oct. 19, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: H10F39/00 (20250101); H04N25/531 (20230101); H04N25/79 (20230101)

U.S. Cl.:

CPC H10F39/8057 (20250101); H04N25/79 (20230101); H10F39/024 (20250101);
H04N25/531 (20230101); H10F39/8027 (20250101); H10F39/8037 (20250101)

Field of Classification Search

CPC: H01L (27/14623); H01L (27/14685); H01L (27/14607); H01L (27/14612); H01L
(27/14609); H01L (27/14643); H01L (27/14683); H04N (25/75); H04N (25/79); H04N
(25/531)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2010/0285630	12/2009	Lee	N/A	N/A
2013/0299931	12/2012	Chuang et al.	N/A	N/A
2015/0221547	12/2014	Arnold et al.	N/A	N/A
2016/0111461	12/2015	Ahn et al.	N/A	N/A
2016/0204150	12/2015	Oh	257/229	H01L 27/14623
2016/0343756	12/2015	Fan	N/A	H01L 27/1462
2017/0005121	12/2016	Lenchenkov et al.	N/A	N/A
2017/0194195	12/2016	Cohen et al.	N/A	N/A
2018/0286905	12/2017	Tsao et al.	N/A	N/A
2019/0393255	12/2018	Lee	N/A	N/A
2020/0006435	12/2019	Hasegawa	N/A	H10K 30/353
2020/0195869	12/2019	Hatano et al.	N/A	N/A
2020/0295068	12/2019	Sato	N/A	H01L 27/14623
2021/0020677	12/2020	Togo et al.	N/A	N/A
2021/0130944	12/2020	Holmes et al.	N/A	N/A
2021/0134869	12/2020	Jang	N/A	H10F 39/807

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202015225	12/2019	TW	N/A
202029484	12/2019	TW	N/A

Primary Examiner: Haliyur; Padma

Attorney, Agent or Firm: Lippes Mathias LLP

Background/Summary

(1) This application is a divisional of U.S. Ser. No. 17/382,571 filed Jul. 22, 2021 and now issued as U.S. patent Ser. No. 11/728,362, which claims the benefit of U.S. provisional application Ser. No. 63/178,398 filed Apr. 22, 2021.

BACKGROUND

(1) The following relates to image sensor pixels and methods of manufacturing image sensor pixels, and to image sensors comprising such image sensor pixels, and to related arts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1(A) diagrammatically illustrates an image sensor and an image sensor pixel of the image sensor according to some embodiments disclosed herein, and FIG. 1(B) diagrammatically illustrates a simplified physical layout of the photodetector and charge storage device of the image sensor pixel of FIG. 1(A).

(3) FIG. 2 diagrammatically illustrates an image acquisition method suitably performed by an image sensor comprising an array of image sensor pixels as set forth in FIGS. 1(A) and 1(B).

(4) FIG. 3 diagrammatically illustrates an image sensor pixel according to an illustrative embodiment.

(5) FIGS. 4(A), 4(B), 4(C), and 4(D) diagrammatically illustrate successive operations in a method of manufacturing the image sensor pixel of FIG. 3 in accordance with some fabrication method embodiments.

(6) FIG. 5 diagrammatically illustrates gas flow during the operation of FIG. 4(C) in one illustrative embodiment in which the light blocking material comprises a Ti/TiN/W multilayer.

(7) FIGS. 6(A) and 6(B) diagrammatically illustrate a poor Ti/TiN/W multilayer deposition corresponding to the abnormal gas flow example of FIG. 5.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or

at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) With reference to FIGS. 1A, an image sensor **10** comprises an array of image sensor pixels **12**, of which an electrical schematic of one image sensor pixel **12** is shown in the main drawing of FIG. 1A. The illustrative image sensor pixel **12** includes a photosensitive device **14**, a charge storage device **16**, and a readout circuit **18**. The readout circuit **18** is also sometimes referred to as a driving circuit **18**. The photosensitive device **14** may, for example, comprise an illustrative photodiode **14** such as a P-N junction photodiode formed of a first doped region **20** of N-type and a second doped region **22** of P-type (or vice versa). The illustrative photodiode is merely an example, and other types of photosensitive devices can be used, such as a phototransistor. The illustrative charge storage device **16** also comprises a P-N junction formed of a first doped region **24** of N-type and a second doped region **26** of P-type (or vice versa, i.e. the photodiode may in general be an N-P diode or a P-N diode). In some embodiments, the first doped regions **20**, **24** of the respective photodiode **14** and charge storage device **16** are formed simultaneously, for example in an ion implantation or dopant diffusion step or by an epitaxial deposition, and likewise the second doped regions **22**, **26** of the respective photodiode **14** and charge storage device **16** may be formed simultaneously. However, this is merely an illustrative example, and in other embodiments the first doped regions **20**, **24** are formed in separate operations and likewise the second doped regions **22**, **26**.

(11) FIG. 1(B) shows another diagrammatic representation of the image sensor pixel **12**, also depicting the photosensitive device **14** and the charge storage device **16**, but omitting the readout circuit **18** and some other electronic components in FIG. 1(B) to focus on the basic pixel layout. As seen in FIG. 1(B), the image sensor pixel **12** is fabricated on and/or in a semiconductor base material **28**, such as an illustrative silicon substrate or layer **28**. In some embodiments, the image sensor pixel **12** is manufactured in complementary metal-oxide-semiconductor (CMOS) technology on the base silicon wafer or substate **28**, although other designs such as a charge-coupled device (CCD)-based image sensor, or an image sensor formed on a GaAs or other type of base semiconductor, are contemplated. In a typical layout, the image sensor **10** is fabricated on a silicon wafer or other semiconductor wafer, and while the illustrative image sensor **10** of FIG. 1(A) comprises a 16×16 pixel array more generally the image sensor may include a much larger pixel array, e.g. on the order of megapixels for some image sensors used in digital cameras or industrial imaging devices. Moreover, the pixels may be arranged over a different type of geometrical area than the illustrative square area, and/or the pixels may optionally be arranged differently than the regular rows and columns of pixels illustrated in FIG. 1(A). In the illustrative example of FIG. 1B, isolation regions **30**, e.g. fabricated as shallow trench isolation (STI) regions, provides isolation between adjacent pixels **12** of the image sensor **10**.

(12) With continuing reference to FIGS. 1(A) and 1(B) and with further reference to FIG. 2, a typical image acquisition sequence includes an operation S1 which starts an exposure of the image sensor by starting the measurement of light by simultaneously resetting all image sensor pixels **12** of the image sensor **10** using a global shutter signal conveyed to each pixel **12** via a shutter gate transistor **32**. In an operation S2, during an exposure time interval (e.g., set by a shutter speed in the case of a digital camera), photocharge is accumulated at each pixel **12**, and more particularly at the photodiode **14** of each pixel **12**. At the end of the exposure, in an operation S3 the accumulated photocharge of the photodiode **14** of each pixel **12** is transferred to the charge storage device **16** of that pixel **12** via a first transfer gate transistor **34**. In an operation S4, the readout circuit **18** of each pixel **12** reads out the charge stored in the charge storage device **16** via a second transfer gate transistor **36**. Advantageously, since the accumulated photocharge is transferred to the charge storage device **16**, the image is fixed at the end of operation S3 and the readout operation S4 can take additional time or process some pixels sequentially, e.g. by reading out charge from the charge storage devices **16** of a row of pixels sequentially in one approach. It is to be appreciated that the

illustrative readout circuit **18** is merely a diagrammatic example, and the detailed readout circuitry may vary depending upon the design of the image sensor **10**. Moreover, the image sensor **10** may include numerous other components which are not shown, such as red, green, and blue color filters for pixels in the case of a full-color image sensor, a mechanical shutter to augment the described electronic shuttering, on-chip exposure metering and/or image processing circuitry, and/or so forth. (13) In some nonlimiting illustrative embodiments, the image sensor **10** is implemented as a CMOS image sensor with global shutter. In this embodiment, the image acquisition is performed as described with reference to FIG. 2, and the operations S1 and S3 implement the global shutter by ensuring the exposure of all pixels **12** of the image sensor **10** start and stop the photocharge accumulation simultaneously. In other embodiments, the image sensor **10** may employ another type of shuttering implementation such as a rolling shutter in which there may be a time delay between reset and readout of successive rows of pixels, for example.

(14) Regardless of the detailed design and shuttering implementation, the charge storage device **16** and associated electronics (e.g., first transfer gate **34**) provide a way to precisely stop the light exposure and temporarily store the photocharge accumulated in the photodiode **14**, without further charge accumulation, to allow for the readout to be subsequently performed. To provide a compact layout for the image sensor **10**, and to minimize any loss of photocharge during the charge transfer, the charge storage device **16** of each pixel **12** is physically located close to the photodiode **14** in the device layout, e.g. both disposed within a designated pixel area. However, if the charge storage device **16** is of a type that is sensitive to light, then any light exposure to the light storage device **16** would result in a measured accumulated charge that is larger than the photocharge accumulated by the photodiode **14** during the exposure operation S2. As previously noted, in some embodiments the doped regions **20**, **24** of the respective photodiode **14** and charge storage device **16** are formed simultaneously, and likewise the second doped regions **22**, **26** of the respective photodiode **14** and charge storage device **16** are formed simultaneously. Hence, in these embodiments the charge storage device **16** has comparable light sensitivity to the photodiode **14**. Even if a different fabrication process is used for the charge storage device **16**, typically any P-N junction and many transistor designs are photosensitive and will accumulate photocharge when exposed to light.

(15) Hence, to prevent the charge storage device **16** from being exposed to light, a light blocking shield or layer **40** is disposed over the charge storage device **16**. Formation of the light blocking shield or layer **40** can be challenging, however. Notably, an electrical path **42** to the charge storage device **16** should be provided, in addition to the light blocking layer **40**, as diagrammatically shown in FIG. 1(A).

(16) With reference now to FIG. 3, an implementation of the light blocking layer **40** and electrical path **42** according to one embodiment is described. In this embodiment, the electrical path **42** is implemented as a light blocking plug **42**. Advantageously, the light blocking layer **40** is not separated from the light blocking plug **42** by a gap. This is in contrast to in some other designs, in which an annular gap may be surrounding the light blocking plug. The light blocking layer **40** and light blocking plug **42** are suitably made of a metal or other light absorbing material that is absorbing for light L that is detected by the photosensitive device **14**, as diagrammatically shown in FIG. 3. As will be described (see FIGS. 4A-4D and related discussion), the light blocking layer **40** and the light blocking plug **42** may be formed together as a single unitary structure. Accordingly, in some embodiments the light blocking layer **40** and the light blocking plug **42** may be made of the same material.

(17) In one nonlimiting illustrative example, the light blocking layer **40** in FIG. 3 may comprise a metal, such as a titanium/titanium nitride/tungsten (Ti/TiN/W) multilayer. A dielectric layer **44** is disposed over the light blocking layer **40**, and the light blocking plug **42** passes through the ILD layer **44** to provide electrical access to the charge storage device **16**. The dielectric layer **44** is commonly referred to as an interlayer dielectric (ILD) layer **44**, as it typically serves to separate different functional and/or interconnect layers of the overall image sensor **10**. Optionally, at least

one contact etch stop layer (CESL) **46** may be deposited for use as an etch stop in certain process steps. In FIG. **3**, the light blocking plug **42** also passes through the CESL **46** to directly contact the underlying charge storage device **16**; however, if the CESL **46** is sufficiently thin then in some embodiments the light blocking plug **42** may make electrical contact through the CESL **46**.

(18) As noted, in the embodiment of FIG. **3**, the light blocking layer **40** is not separated from the light blocking plug **42** by any gap. Rather, the light blocking layer **40** and light blocking plug **42** are a contiguous structure made of a single material. For example, in one embodiment the single material comprises a Ti/TiN/W multilayer.

(19) The implementation of the light blocking layer **40** and light blocking plug **42** of FIG. **3** has certain advantages. One advantage is that the approach of FIG. **3** eliminates having a gap between the light blocking plug **42** and the light blocking layer **40**. Such a gap would present a light leakage path by which a portion of the light **L** can impinge on the underlying charge storage device **16**. This can cause undesired additional photocharge forming in the charge storage device **16**. For example, referring to FIG. **2**, during the exposure time of the step **S2** there should ideally be no charge building up in the charge storage device **16**. However, if there were light leakage through a gap between the light blocking layer **40** and the light blocking plug **42**, then some charge would build up in the charge storage device **16** during the step **S2**. When, in the step **S3**, the photocharge accumulated in the photosensitive device **14** is transferred to the charge storage device **16**, this would result in an excess amount of charge being stored in the charge storage device **16**—namely, the charge stored will be the photocharge accumulated in the photosensitive device **14** over the exposure time plus the photocharge accumulated in the charge storage device **16** over the exposure time due to light leakage through such gap between the light blocking layer and the plug.

Furthermore, unless a mechanical shutter physically blocks light **L** from reaching the pixel **12**, the photocharge in the charge storage device **16** would continue to increase over the course of the subsequent readout operation **S4** due to the continuing light leakage through such a gap.

(20) By contrast, in the embodiment of FIG. **3**, there is no gap between the light blocking layer **40** and light blocking plug **42**, because in the embodiment of FIG. **3** the light blocking layer **40** and light blocking plug **42** are a contiguous structure with no gap therebetween. This eliminates the undesirable accumulation of charge in the charge storage device **16** during the operation **S2** (and also during the operation **S4** if no mechanical shutter physically blocks the light **L** during the operation **S4**).

(21) The skilled artisan might initially consider that having the light blocking layer **40** and light blocking plug **42** in contact with no gap therebetween might adversely impact electrical operation of the charge storage device **16**, since if the single material making up the contiguous structure is a metal or other electrically conductive material then the light blocking layer **40** is in galvanic electrical contact with the light blocking plug **42** that provides electrical access to the charge storage device **16**. However, it is recognized herein that the light blocking layer **40** is electrically isolated by the ILD layer **44** (and optionally also by the underlying CESL **46** and/or silicon substrate or layer **28**). Hence, this electrical contact between the light blocking layer **40** and light blocking plug **42** is not problematic.

(22) A further advantage of the implementation of the light blocking layer **40** and light blocking plug **42** of FIG. **3** is that the implementation of FIG. **3** can be efficiently fabricated, as it requires fewer fabrication steps than an implementation in which the light blocking layer **40** and light blocking plug **42** are formed separately.

(23) With reference to FIGS. **4(A)**-**4(D)**, an embodiment of a fabrication process for fabricating the light blocking layer **40** and light blocking plug **42** of FIG. **3** is described. As the focus of this fabrication process is on the light blocking components **40**, **42**, FIGS. **4(A)**-**4(D)** illustrate the charge storage device **16** without showing the photosensitive device **14** or other components of the image sensor pixel **12**. FIG. **4(A)** depicts the charge storage device **16** on the semiconductor base material **28**, with the CESL **46** and ILD layer **44** already deposited. The fabrication of these

features is not illustrated. In general, the charge storage device **16** may be fabricated on the semiconductor base material **28** using any suitable approach for the type of charge storage device employed, for example ion implantation, dopant diffusion, epitaxial deposition, or so forth to form the doped regions **24**, **26** (see FIG. **1(A)**) and a metal or other electrically conductive layer deposition to form a gate electrode. The optional CESL **46** may comprise, by way of nonlimiting illustrative example, silicon nitride, carbon-doped silicon nitride, or a combination thereof. The optional CESL **46** may be deposited using, for example, chemical vapor deposition (CVD), high density plasma (HDP) CVD, sub-atmospheric CVD (SACVD), molecular layer deposition (MLD), or other suitable methods, for use as an etch stop in certain process steps. The ILD **44** may comprise silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), spin-on glass (SOG), fluorinated silica glass (FSG), carbon doped silicon oxide (e.g., SiCOH), polyimide, a low-k dielectric material, such as Xerogel, Aerogel, amorphous fluorinated carbon, Parylene, BCB (bis-benzocyclobutenes), hydrogen silsesquioxane (HSQ) or fluorinated silicon oxide (SiOF), and/or various combinations thereof. In some embodiments, the ILD **44** is formed to a suitable thickness by Flowable CVD (FCVD), CVD, HDPCVD, SACVD, spin-on, sputtering, or other suitable methods.

(24) In the process of FIGS. **4(A)**-**4(D)**, the ILD layer **44** is formed as a first ILD layer **44.sub.1** and a second ILD layer **44.sub.2** that is deposited on top of the first ILD layer **44.sub.1**. The second ILD **44.sub.2** is deposited prior to any etching for formation of the light blocking layer **40** or light blocking plug **42**, as seen in FIG. **4(A)**. In the illustrative example, the first (lower) ILD layer **44.sub.1** and a second (upper) ILD layer **44.sub.2** are made of different materials. As will be described with reference to FIG. **4(C)**, providing the ILD layer **44** as these two different layers **44.sub.1** and **44.sub.2** of different materials will enable selective etching of the first (lower) ILD layer **44.sub.1** so as to undercut the second (upper) ILD layer **44.sub.2** so as to open a space for the subsequent formation of the light blocking layer **40** underneath the second (upper) ILD layer **44.sub.2**. In one nonlimiting illustrative example, the first (lower) ILD layer **44.sub.1** comprises a BPSG layer **4141**, and the second (upper) ILD layer **44.sub.2** comprises a PE-TEOS layer **44.sub.2**. These are merely illustrative examples.

(25) With reference to FIGS. **4(B)** and **4(C)**, a selective etching operation or operations is performed to open a contiguous volume **60**, **62** in the ILD layer **44**. This contiguous volume includes a sub-volume **60**, **62** comprising an undercut volume **60** corresponding to the volume of the light blocking layer **40**, and a sub-volume **62** comprising an access opening corresponding to the volume of the light blocking plug **42**. In the illustrative example, a first etch operation is performed which etches through the upper ILD layer **44.sub.2** (e.g., a PE-TEOS layer in some embodiments), and through the lower ILD layer **44.sub.2** (e.g., a BPSG layer in some embodiments), and the optional CESL **46**, to form an opening **64** that penetrates to the charge storage device **16** as shown in FIG. **4(B)**. This opening **64** corresponds to the access opening **62** and the portion of the sub-volume **60** directly underneath the access opening **62**. In an alternative embodiment, if the CESL **46** is sufficiently thin so that it does not introduce unacceptably high electrical resistance between the light blocking plug **42** and the electrode of the charge storage device **16**, then it is contemplated for the opening **64** to not penetrate through the CESL **46**. Photolithographic patterning may be used to selectively etch to form the access opening **62**. The etchant is suitably chosen based on the type of material(s) making up the lower and upper ILD layers **44.sub.1**, **44.sub.2** and the CESL **46** (if it is to be etched). Notably, the upper portion of the opening **64** corresponds to the sub-volume **62** comprising the access opening corresponding to the volume of the light blocking plug **42**.

(26) Thereafter, as shown in FIG. **4(C)**, a second etch operation is performed to extend the opening **64** formed in the first etch of FIG. **4(B)** to form the undercut volume **60** corresponding to the volume of the light blocking layer **40**. The second etch has high selectivity for etching the lower ILD layer **44.sub.1** (e.g., BPSG) over the upper ILD layer **44.sub.2** (e.g., PE-TEOS), to produce the

undercutting while leaving overhanging ILD regions **68** as indicated in FIG. **4(C)**. These overhanging ILD regions **68** ensure that the subsequently formed light blocking layer **40** will be buried by the ILD layer **44** (and more particularly in this embodiment by the upper ILD layer **44.sub.2**).

(27) In one nonlimiting illustrative embodiment in which the lower ILD layer **44.sub.1** comprises a BPSG and the upper ILD layer **44.sub.2** comprises PE-TEOS, the etching of FIGS. **4(B)** and **4(C)** is performed as follows. A nonselective contact etch (CT etch) in the form of an isotropic dry etch is performed to form the opening **64** shown in FIG. **4(B)**. This nonselective etch is laterally limited to the access opening **62** by lithographic patterning. Thereafter, the photoresist is removed, e.g. by an ash process, and a selective wet etch is performed that selectively etches the BPSG (i.e., lower ILD layer **44i**) without etching the PE-TEOS (i.e., upper layer **44.sub.2**), thereby leaving the overhanging ILD regions **68** of PE-TEOS (i.e., upper layer **44.sub.2**). In one nonlimiting illustrative embodiment, the selective wet etch uses Caro's acid. In some embodiments, the B/P ratio of the BPSG and the etch time are optimized by experimentation to obtain high etch selectivity for the BPSG over the PE-TEOS and the desired undercutting to leave the overhanging ILD regions **68**. If the CESL **46** is provided, then this may optionally serve as an etch stop layer for the selective etching.

(28) More generally, if the ILD layer **44** includes a lower ILD layer **44.sub.1** and an upper ILD layer **44.sub.2** as shown in the example of FIGS. **4(A)**-**4(D)**, then the first etch is a nonselective etch that is designed to etch through the materials of both ILD layers **44.sub.1** and **44.sub.2**, and through the CESL **46** if it is included in the device structure. The second, selective etching process is suitably designed to have high etch selectivity for etching the lower ILD layer **44.sub.1** over the upper ILD layer **44.sub.2**, to produce the desired undercutting forming overhangs **68**. The choice of etchant may be made based on the materials of the lower and upper ILD layers **44.sub.1**, **44.sub.2**.

(29) It will be further appreciated that the ILD layer **44** may similarly comprise three (or more) ILD layers of different materials. In these embodiments, the nonselective first etch (analogous to FIG. **4(B)**) should be chosen to etch through all the materials of the multilayer ILD **44** to expose access to the charge storage device **16**, and the second, selective etch (analogous to FIG. **4(C)**) should selectively etch one or more, but not all, of the materials of the multilayer ILD **44** to form the undercut volume **60** while leaving the overhangs **68**.

(30) Moreover, it is contemplated the approach may also be employed in the case of an ILD layer **44** comprising a single layer made of a single material. In this case the etching is designed to have sufficient anisotropy to preferentially etch downward to form the access opening **62** and penetrate down to the charge storage device **16**, but to also provide some lateral etching to form the undercut volume **60**.

(31) While FIG. **4(C)** shows the contiguous volume **60**, **62** with idealized straight edges and right-angle corners, in practice the edges may be curved and/or the corners may be substantially rounded, or the shape may otherwise deviate from the idealized diagrammatic representation of FIG. **4(C)**. For example, it is sufficient that the undercut volume **60** be of sufficient lateral extent to be able to be filled to form the light blocking layer **40** completely covering the charge storage device **16**, and of sufficient vertical dimension so that the subsequently formed light blocking layer **40** has sufficient thickness to provide substantially complete light absorption to avoid photoinduced charge from being optically injected into the charge storage device **16**.

(32) With reference to FIG. **4(D)**, the contiguous opening **60**, **62** formed by the selective etching described with reference to FIGS. **4(B)** and **4(C)** is filled in with a metal or other light blocking material to form both the light blocking layer **40** and the light blocking plug **42**. Put another way, a light blocking material is deposited to both: (i) fill the undercut volume **60** with the light blocking material to form a light blocking layer **40** covering the charge storage device **16**, and (ii) fill the access opening **62** with the light blocking material to form a light blocking plug **42**. These filling operations (i) and (ii) may be performed simultaneously, for example using a physical vapor

deposition (PVD) technique. The PVD fills the contiguous volume **60, 62** with the light blocking material to form both the light blocking layer **40** and the light blocking plug **42**. This forms the light blocking layer **40** and the light blocking plug **42** as a contiguous structure **40, 42** made of the light blocking material, and hence the light blocking layer **40** and the light blocking plug **42** are made of the same light blocking material. The PVD may consist of a single deposition operation, although it will be appreciated that gas flows may be varied, possibly independently, as a function of time during this single deposition operation. For example, in one nonlimiting illustrative embodiment the light blocking material is a titanium/titanium nitride/tungsten (Ti/TiN/W) multilayer formed by varying the flows of triethylborate (TEB), tetraethoxysilane (TEOS), and triethylphosphate (TEPO). The choice of PVD technique is suitably chosen based on factors such as the type of light blocking material to be deposited, and providing a deposition rate that is effective to completely fill contiguous volume **60, 62**.

(33) Parameters for the selective etching for forming the contiguous volume **60, 62**, and for the PVD for filling the contiguous volume **60, 62** with the light blocking material to form the light blocking layer **40** and the light blocking plug **42**, can be optimized empirically by forming test structures and assessing the resulting light blocking layer **40** and the light blocking plug **42** using characterization techniques such as transmission electron microscopy (TEM) and electrical characterization of the electrical resistance between the light blocking plug **42** and the charge storage device **16**.

(34) With reference to FIGS. **5, 6A**, and **6B**, a diagrammatic example of such optimization is illustrated. FIG. **6A** depicts a charge storage device **16** over which is disposed the ILD layer **44** including the lower ILD **44.sub.1** which in this embodiment is a BPSG layer, and the upper ILD **44.sub.2** which in this embodiment is a PE-TEOS layer. The illustrative example of FIGS. **6(A)** and **6(B)** further includes an anti-reflection coating (ARC) **80**. FIG. **6A** further diagrammatically depicts a selectively etched volume **82** which corresponds to the contiguous volume **60, 62** of previous examples. Arrows **84** indicate the gas flow during PVD of the material forming the light blocking layer **40** and light blocking plug **42** into the etched volume **82** (e.g., corresponding to the operation described previously with reference to FIG. **4(D)**). FIG. **5** plots the gas flows of TEB, TEOS, and TEPO during a normal run (top plot) and an abnormal run (bottom plot). FIG. **6B** illustrates how during an abnormal run the resulting Ti/TiN/W multilayer **86** (corresponding to the contiguous structure **40, 42** made of the light blocking material of previous embodiments) does not fully fill the selectively etched volume **82**, but instead leaves a void **88**. The void can adversely impact device yield. To optimize the process, multiple runs can be performed with different gas flow settings (e.g., flow ramp start, ramp rate, et cetera) and the resulting devices characterized by TEM to determine the gas flow recipe that optimally fills the selectively etched volume **82**.

(35) In the following, some additional embodiments are disclosed.

(36) In an illustrative embodiment, a method is disclosed for providing light shielding of a charge storage device of an image sensor pixel. The method includes providing a photosensitive device and the charge storage device and a dielectric layer covering the photosensitive device and the charge storage device. The method further includes performing etching of the dielectric layer to define an undercut volume beneath the dielectric layer and an access opening through the dielectric layer to the undercut volume, and performing physical vapor deposition (PVD) of a light blocking material to both: fill the undercut volume with the light blocking material to form a light blocking layer covering the charge storage device, and fill the access opening with the light blocking material to form a light blocking plug.

(37) In another illustrative embodiment, an image sensor pixel includes a photosensitive device, a charge storage device, a light blocking layer disposed over the charge storage device, a light blocking plug disposed over the charge storage device, and a dielectric layer disposed over the light blocking layer. The light blocking plug passes through the dielectric layer and contacts the light blocking layer.

(38) In another illustrative embodiment, an image sensor comprises an array of image sensor pixels as set forth in the immediately preceding paragraph.

(39) In another illustrative embodiment, a method is disclosed for providing light shielding of a charge storage device of an image sensor pixel that includes a photosensitive device and the charge storage device and a first dielectric layer made of a first material and covering the photosensitive device and the charge storage device and a second dielectric layer made of a second material different from the first material and covering the first dielectric layer. The method includes: performing a nonselective etch through the second dielectric layer to define an access opening through the second dielectric layer and through the first dielectric layer to expose the charge storage device; after the nonselective etch, performing a selective etch that selectively etches the first material over the second material to define an undercut volume laterally extending at least over the charge storage device; and after the nonselective etch, performing physical vapor deposition (PVD) of a light blocking material to fill the access opening and the undercut volume to form a light blocking layer covering the charge storage device and a light blocking plug.

(40) In another illustrative embodiment, an apparatus includes a charge storage device, a light blocking layer, a dielectric layer, and a light blocking plug. The light blocking layer is disposed over the charge storage device. The dielectric layer is disposed over the light blocking layer. The light blocking plug is disposed over the charge storage device and passes through the dielectric layer to provide electrical access to the charge storage device. The light blocking layer and the light blocking plug comprise a contiguous structure. In some embodiments, there is no gap between the light blocking layer and the light blocking plug. In some illustrative examples, the contiguous structure may be made of a metal or of a multilayer.

(41) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An image sensor pixel comprising: a photosensitive device; a charge storage device; a charge transfer path that transfers electrical charge from the photosensitive device to the charge storage device; a light blocking layer disposed over the charge storage device; a light blocking plug disposed over the charge storage device; and a dielectric layer disposed over the light blocking layer; wherein the light blocking plug passes through the dielectric layer and contacts the light blocking layer, and also electrically contacts the charge storage device.
2. The image sensor pixel of claim 1 wherein there is no gap between the light blocking layer and the light blocking plug.
3. The image sensor pixel of claim 1 wherein the light blocking layer and the light blocking plug comprise a contiguous structure.
4. The image sensor pixel of claim 3 wherein the contiguous structure is made of a metal or of a multilayer.
5. The image sensor pixel of claim 1 further comprising: a shutter gate transistor connected to reset the photosensitive device; readout circuitry configured to read out electrical charge stored in the charge storage device; and wherein the charge transfer path includes a transfer gate transistor via.
6. An image sensor comprising an array of image sensor pixels as set forth in claim 5.
7. The image sensor pixel of claim 1 wherein the dielectric layer includes: a first dielectric layer of

a first material proximate to the charge storage device, and a second dielectric layer of a second material distal from the charge storage device, the second dielectric material being different from the first dielectric material.

8. The image sensor pixel of claim 7 wherein: the light blocking layer is disposed in the first dielectric layer, the second dielectric layer is disposed over the light blocking layer and the first dielectric layer, and the light blocking plug passes through the second dielectric layer and contacts the light blocking layer.

9. The image sensor pixel of claim 1 wherein the dielectric layer includes: a borophosphosilicate glass (BPSG) layer proximate to the charge storage device, and a plasma enhanced tetraethoxysilane (PE-TEOS) layer distal from the charge storage device.

10. The image sensor pixel of claim 9 wherein: the light blocking layer is disposed in the BPSG layer, the PE-TEOS layer is disposed over the light blocking layer and the BPSG layer, and the light blocking plug passes through the PE-TEOS layer and contacts the light blocking layer.

11. An image sensor pixel comprising: a charge storage device; a charge transfer path that transfers electrical charge from a photosensitive device to the charge storage device; a light blocking layer disposed over the charge storage device; a dielectric layer disposed over the light blocking layer; and a light blocking plug disposed over the charge storage device and passing through the dielectric layer to provide electrical access to the charge storage device; wherein the light blocking layer and the light blocking plug comprise a contiguous structure.

12. The image sensor pixel of claim 11 wherein there is no gap between the light blocking layer and the light blocking plug.

13. The image sensor pixel of claim 11 wherein the contiguous structure is made of a metal or of a multilayer.

14. The image sensor pixel of claim 11 wherein the dielectric layer includes: a first dielectric layer of a first material proximate to the charge storage device, and a second dielectric layer of a second material distal from the charge storage device, the second dielectric material being different from the first dielectric material.

15. The image sensor pixel of claim 14 wherein: the light blocking layer is disposed in the first dielectric layer, the second dielectric layer is disposed over the light blocking layer and the first dielectric layer, and the light blocking plug passes through the second dielectric layer.

16. The image sensor pixel of claim 14 wherein the dielectric layer includes: a borophosphosilicate glass (BPSG) layer proximate to the charge storage device, and a plasma enhanced tetraethoxysilane (PE-TEOS) layer distal from the charge storage device.

17. The image sensor pixel of claim 16 wherein: the light blocking layer is disposed in the BPSG layer, the PE-TEOS layer is disposed over the light blocking layer and the BPSG layer, and the light blocking plug passes through the PE-TEOS layer.

18. An image sensor comprising: a charge storage device; a charge transfer path that transfers electrical charge from a photosensitive device to the charge storage device; a first dielectric layer disposed over the charge storage device; a second dielectric layer disposed over the first dielectric layer, the second dielectric layer comprising a different material than the first dielectric layer; and a structure comprising an electrically conductive material, the structure electrically contacting the charge storage device, the structure including a light blocking shield disposed in the first dielectric layer and disposed over the charge storage device.

19. The image sensor of claim 18, wherein the second dielectric layer is disposed over the light blocking shield.

20. The image sensor of claim 19, wherein the first dielectric layer is a borophosphosilicate glass (BPSG) layer and the second dielectric layer is a plasma enhanced tetraethoxysilane (PE-TEOS) layer.
